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Potential: “intrinsic energy of inertia”

Force caused by Potential: Electro-Magnetic Field

- static: localized source charge at rest, only spacial dependency, no time dependency
- stationary: source charge at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistic: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movement
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(cp_{0_{dipol}} + m_0 \right) \quad (1)$$

Magnetic Field in Matter

Excitation $\hat{=}$ Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservative}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} \cdot \frac{v}{c}\right)\right]}{c}}_{\text{acceleration}} \right] \quad (3)$$

Electric Field in Matter

Displacement $\hat{=}$ Excitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Excitation: } \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} = \overset{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \underset{(4)}{\varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n}$$

Abbreviations

Φ = electric potential A = magnetic potential $j = \nabla \times \mathcal{H}$ = elec. current density

$\mathcal{E}$ = Electric Field $\mathcal{B}$ = Magnetic Field $\nabla \cdot \mathcal{B} = 0$ $\mathcal{D}$ = Displacement density of dielectricum $\mathcal{M} =$

$\mathcal{P}$ = Polarisation $\mathcal{H}$ = magn. excitation, magn. field strength ("magn. displacement")

n = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$ χ = elec. susceptibility of dielectricum $\kappa = \frac{\mu - \mu_0}{\mu_0} = \text{magn. susceptibility}$

$\langle p_m \rangle$ = spacial average of magn. dipol moment $\langle \mathcal{E}_m \rangle$ = spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field μ = magn. permeability μ_0 = magn. constant ...

ε = dielectricity constant $\varepsilon_0 = \dots$

Interdependencies of Fields: Electro-Magnetic System (Maxwell conditions)

Static, kinetic rest charge localisation (conservative system):

...

Stationary, slow charge movement localisation (conservative system):

$$\nabla \cdot \varepsilon_0 \mathcal{E} = \varrho \quad \nabla \cdot \mathcal{B} = 0 \quad \nabla \times \mathcal{E} = -\frac{\partial}{\partial t} \mathcal{B} = i\omega \mathcal{B} \quad \nabla \times \frac{1}{\mu_0} \mathcal{B} = j + \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} = j - i\omega \varepsilon_0 \mathcal{E} \quad (5)$$

Dynamic, fast charge movement (relativistic):

...

Potential derivation

⇒ Wave equation that determines Potential:

$$\Delta \cdot U - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} U = J \quad (6)$$

⇒ Potential caused by “ignition J” (inhomogeneity) from source at rest, $v = 0$:

$$U = -\frac{1}{4\pi} \int \frac{J}{|r|} dr' \quad (7)$$

elec. Potential structure

$$J = -\frac{1}{\varepsilon_0} \varrho$$

$$U_e := \Phi \stackrel{\text{elec. charge dens.}=\varrho}{=} \frac{1}{4\pi} \frac{1}{\varepsilon_0} \int \frac{\varrho}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} Q = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{\varepsilon_0} It \quad (8)$$

magn. Potential structure

$$J = -\mu_0 j$$

$$U_m := A \stackrel{\text{elec. current dens.}=j}{=} \frac{\mu_0}{4\pi} \int \frac{j}{|r|} dr' \hat{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \mu_0 I = \frac{1}{2} \frac{1}{2\pi} \frac{1}{|r|} \frac{1}{c^2 \varepsilon_0} I = \frac{1}{c^2} \frac{1}{t} U_e \stackrel{\text{source in uniformal movement, } v=\text{const}}{=} \quad (9)$$

Electric endogen energy

- Bond Energy
- Binding Energy

$$E_{pot} = mc^2$$

$$e = \text{electron charge} \quad q = ne = \text{charge} \quad n \in \mathbb{N} \quad (10)$$

$$\text{Potential} \quad U = \frac{E_{pot}}{q} = \frac{(mc^2)^2}{ne} = \frac{m}{q} mc^4 = \kappa mc^4 \quad (11)$$

Electric induced energy (exogenous excitation)

- Ionisation Energy
- Excitation Energy

$$E_{kin} = \frac{1}{2} mv^2 \hat{=} (pc)^2 \hat{=} h\nu = \hbar\omega$$

E_{kin} Energy density electric:

$$w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2 \quad (12)$$

E_{kin} Energy density magnetic:

$$w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \quad (13)$$

E_{kin} Electric Energy:

$$W_e = \frac{1}{2} CU^2 \quad (14)$$

E_{kin} Magnetic Energy:

$$W_m = \frac{1}{2} LI^2 \quad (15)$$

Power:

$$P = VI = (U_1 - U_2) \frac{Q}{t} = \frac{E_1 - E_2}{t} = \frac{W}{t} = \frac{Fr}{t} = Fv \quad (16)$$

Frequency:

$$\omega = \dot{\rho} = \frac{v}{r} = 2\pi\nu = k\lambda \frac{1}{T} = \sqrt{\frac{k}{m}} \stackrel{vacuum}{=} kc \stackrel{matter}{=} k \frac{c}{\eta} = kc \frac{\tan \beta}{\tan \alpha} \stackrel{electromagnetic}{=} kc \left(\mathcal{E}_\beta^{-1} \mathcal{E}_\alpha \right)^{-1} = kc \frac{\varepsilon_\beta}{\varepsilon_\alpha} = kc \sqrt{\frac{\varepsilon_0 \mu}{\varepsilon \mu}} \quad (17)$$

Phase:

$$\varphi = k \left(\vec{r} \cdot \vec{k} \right) - \omega t := ks - \omega t = \dots \quad (18)$$

Phaseshift:

$$\Delta\varphi = \Delta k \left(\vec{r} \cdot \vec{\Delta k} \right) - \Delta\omega t := \Delta ks - \Delta\omega t = \dots \quad (19)$$

Refraction:

Refraction $\Leftrightarrow \exists$ Matter; refraction causes dispersion

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\lambda_0}{\lambda} = c\sqrt{\varepsilon\mu} \hat{=} \mathcal{E}_\beta^{-1} \mathcal{E}_\alpha = \frac{\varepsilon_\alpha}{\varepsilon_\beta} = \dots \quad (20)$$

Dispersion:

Dispersion $\Leftrightarrow \exists$ Matter (dielectric medium or conductor); Dispersion $\neq$ Dissipation $\neq$ Diffusion

$$\dots \quad (21)$$

Dielectricity:

$$\varepsilon(\omega) = \dots \quad (22)$$

Permeability:

$$\mu(\omega) = \dots \quad (23)$$

Conductivity:

$$\sigma(\omega) = \dots \quad (24)$$

Conductor (charge conductive element for energy transfer) condition:

$$\sigma \cdot t \gg \varepsilon_0 \quad (25)$$

Slow changing current (quasi-stationary, low frequencies) condition:

$$\frac{v}{c} \ll \frac{1}{c} \frac{\mathcal{E}}{\mathcal{B}} \ll \frac{c}{v} \quad (26)$$

Energy of thin conductor (concentrated conductive medium without electric charge in its surrounding), Voltage (potential difference, tension):

$$V = U_1 - U_2 = r\mathcal{E} = \dot{\Psi} + RI + \frac{I}{C}t = L\dot{I} + RI + \frac{Q}{C} \quad (27)$$

Low Frequency

$$\begin{aligned} \omega \ll \frac{1}{\mu\sigma} \frac{2}{r^2} &\iff r \ll d \iff \rho \ll 1 \\ L \approx \text{const.} \quad C \approx \text{const.} \quad D_S \approx 0 \end{aligned} \quad (28)$$

High Frequency

$$\begin{aligned} \omega \gg \frac{1}{\mu\sigma} \frac{2}{r^2} &\iff r \gg d \iff \rho \gg 1 \\ L(t) \neq \text{const.} \quad C(t) \neq \text{const.} \quad \text{for Dipol: } D_S \propto \omega^4 \gg 0 \end{aligned} \quad (29)$$

Continuity at Rest: Electric Charge

no Dissipation, no velocity, charge (ion) causes Electric Field

Continuity at Movement: Magnetic Field

no Dissipation, Velocity constant, moved charge (ion) causes emergence of Magnetic Field

Continuity at Acceleration: Radiation Energy

no Dissipation, Velocity changing, acceleration of charge (ion) causes emergence of Radiation

Continuity and periodic (harmoneous oscillation) Movement in VACUUM

$$c = \frac{1}{\sqrt{\varepsilon_0 \mu_0}} = \nu \lambda = \frac{\lambda}{T}$$

$$2\pi = k\lambda = T\omega = \frac{\omega}{\nu}$$

$$2\pi = k\lambda \gg kr_0 \ll 1$$

$$\omega = kc \ll \frac{c}{r_0} \quad \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Continuity and periodic (harmoneous oscillation) Movement in MATTER

$$\text{Refraction index } \eta \text{ emerges: } \varepsilon\mu = \left(\frac{\eta}{v_{ph}}\right)^2 \hat{=} \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon\mu\omega^2 + i\mu\sigma\omega$$

Near Field

static, vacuum, granular spacetime

$$r \ll \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \ll 2\pi$$

$$\frac{w_m}{w_e} \sim \frac{\frac{\vec{p}^2}{\varepsilon_0 r^6}}{\frac{k^2 \mu_0 c^2 \vec{p}^2}{r^4}} \sim (kr)^2 \ll 1$$

$$\vec{p} = \text{elec. Dipol}$$

Far Field

static, vacuum, flat spacetime

accelerated movement -> Radiation in Far Field of charge source

$$r \gg \lambda = \frac{2\pi}{k} \stackrel{\text{vacuum}}{=} \frac{2\pi}{\frac{\omega}{c}} \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1$$

$$\frac{w_m}{w_e} = \frac{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2}{\frac{\varepsilon_0}{2} c^2 \left(\mathcal{B} \times \frac{r}{|r|} \right)^2} = 1$$

Dissipation in Vacuum

no Continuity of charge/matter, no Refraction index, Reflection

Dissipation in Matter

no Continuity of charge/matter with dielectricum, refraction merges, reflection, absorption

Energy gain by excitation

Action Quantum, Planck Constant (Energy-Frequency-Slope)¹:

$$h = \tan(\varphi) = \frac{E_{kin}}{\nu_i - \nu_0} = \frac{eV_{0i}}{\nu_i - \nu_0} \equiv \frac{E_1 - E_2}{\nu_0} = \frac{W}{\nu_0} = \frac{F_{EM} \cdot x}{\nu_0} = \hbar k \lambda \quad (30)$$

$$\hbar = \frac{E_1 - E_2}{\omega} = \quad (31)$$

$$E_{kin}^{max} = eV_0 = e \frac{Q}{C} = h(\nu - \nu_0) \quad (32)$$

¹Wirkungsquantum

$V_0 =$ Opposing Potential ($I = 0$)

$\varphi = \angle(E_{kin}, \nu)$ $e =$ electron charge $Q =$ ion charge

$$C = \text{Capacitance} \quad \nu = \frac{1}{2\pi} \frac{1}{\sqrt{LC}} \sqrt{1 - \frac{C}{L} R^2}$$

$$\Rightarrow \text{Inductance } L = \frac{4\pi^2}{C} \left(\frac{e}{h} Q + \nu_0 \right)^2$$

Ionisation Energy of chemical element $E_i = h\nu = h \frac{\lambda}{c} = \hbar\omega = \frac{\hbar}{k\lambda}\omega$

Energy released (gained), see... Equation 6

- with mass (e.g. electron), Tunnel-Effect: $E_{gain} = E_{pot} - E_i = mc^2 - h\nu_i$
- massless (e.g. photon), Doppler-Effect (frequency shift): $E_{gain} = E_{kin} - E_i = h(\nu_0 - \nu_i) = \Delta\omega\hbar$

Electromagnetic Energy Exchange in Vacuum

charge conservation:

$$\frac{\partial}{\partial t} \varrho + \nabla \cdot j = 0 \tag{33}$$

$\varrho = \sigma\delta = \varepsilon_0 \nabla \cdot \mathcal{E} =$ charge density in interior

$\sigma =$ charge density at border (skin)

$\delta =$ Dirac delta-function

$j = \sigma(\mathcal{E} + v \times \mathcal{B}) = \nabla \times \mathcal{H} - \frac{\partial}{\partial t} \mathcal{D} = \frac{1}{\mu_0} \nabla \times \mathcal{B} - \varepsilon_0 \frac{\partial}{\partial t} \mathcal{E} \hat{=} \text{elecectric current density}$

$j \cdot \mathcal{E} \hat{=} \text{energy gain by fields } \mathcal{E} \text{ and } \mathcal{B} \text{ through current density } j$

$\mathcal{S} = \mathcal{E} \times \mathcal{H} \hat{=} \text{energy transport density (Intensity} = |\overline{\mathcal{S}}|)$

Energy conservation for (large extended) conductive medium:

$$\int \left(\frac{\partial}{\partial t} (w_e + w_m) + \cdot \mathcal{S} + j \cdot \mathcal{E} \right) dr = \frac{d}{dt} W_e + \frac{d}{dt} W_m + D_S + \int (j \cdot \mathcal{E}) dr \stackrel{\text{low frequency}}{=} \frac{1}{2} \frac{d}{dt} (LI^2) + \frac{1}{2} \frac{d}{dt} \left(\frac{Q^2}{C} \right) \quad (34)$$

Energy Balance

Electro-Magnetic Energy:

$$E = E_{kin} + E_{pot} = \sqrt{(pc)^2 + (mc^2)^2} \stackrel{\text{vacuum}}{=} pc = \hbar\omega = \text{chargeenergy} + \text{fieldenergy} + \text{transportenergy} \quad (35)$$

Charge Energy:

$$\underbrace{j \cdot \mathcal{E}}_{\text{charge energy}} = - \left(\underbrace{\frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} + \underbrace{\nabla \cdot \mathcal{S}}_{\text{transfer energy}} \right) = \mathcal{E} \cdot \nabla \times \mathcal{A} - \mathcal{E} \cdot \frac{\partial}{\partial t} \mathcal{D} \quad (36)$$

Field Energy:

$$\frac{\partial}{\partial t} w = \frac{\partial}{\partial t} \sum w_i = \frac{\partial}{\partial t} (w_e + w_m) = \frac{\partial}{\partial t} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} \mathcal{B}^2 \right) \stackrel{\text{vacuum}}{=} \frac{\partial}{\partial t} \left(\frac{1}{2} \frac{1}{\mu_0} (\mathcal{A} \cdot \nabla \times \mathcal{B} - \mathcal{B}^2) \right) \quad (37)$$

Transfer Energy:

$$-\dot{w} \hat{=} \nabla \cdot \mathcal{S} \stackrel{\text{matter}}{=} \nabla \cdot (\mathcal{E} \times \mathcal{H}) \stackrel{\text{vacuum}}{=} \nabla \cdot (\mathcal{E} \times \mathcal{B}) \stackrel{\text{alternativ in vacuum}}{=} \nabla \cdot \left(\frac{1}{2} \frac{1}{\mu_0} \frac{\partial}{\partial t} (\mathcal{A} \times \mathcal{B}) \right) = \dots \quad (38)$$

Local Energy Conservation:

Force $F = q(\mathcal{E} + v \times \mathcal{B})$

Work $W = \Delta E = Fr = pv = \text{const.}$

Power $\frac{d}{dt} W = 0$

$$\overbrace{j \cdot \mathcal{E} + \frac{\partial}{\partial t} \sum w_i + \sum \mathcal{S}_{w_i}}^{= \text{const.}} \stackrel{\text{alternatives}}{=} j \cdot \mathcal{E} + \frac{\partial}{\partial t} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{A} \cdot \nabla \times \mathcal{B} \right) + \frac{1}{\mu_0} \nabla \cdot \left(\mathcal{E} \times \mathcal{B} + \frac{1}{2} \frac{\partial}{\partial t} \right) \quad (39)$$

charge energy field energy transfer energy

conditions for modeling alternatives (Eigenvalue solution):

- $w = \sum w_i : \frac{\partial}{\partial t} w = -\nabla \cdot \mathcal{S}_w$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $w = w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{E} \lambda^2 = \varrho \lambda$
- $\mathcal{S}_w = \mathcal{S}_w(\mathcal{E}, \mathcal{B}) : \mathcal{B} \lambda^2 = j \lambda$

Energy Density Flux: Poynting-Vector

$$\mathcal{S} \stackrel{\text{vacuum}}{=} \mathcal{E} \times \mathcal{B} \stackrel{\text{matter}}{=} \mathcal{E} \times \mathcal{H}$$

Far Field: Power = $\frac{\Delta E}{t} \hat{=} \frac{d}{dt} E = 0 \Rightarrow \text{Energy} = \text{const.} \Rightarrow \text{Energy Conservation}$

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \underbrace{\left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right)}_{= \text{const.}} = 0$$

Near Field (local): Power = Energytransport

$$\frac{d}{dt} E \hat{=} \frac{d}{dt} \left(\underbrace{\int_{\mathbb{R}^3} j \cdot \mathcal{E}}_{\text{charge energy}} + \underbrace{\int_{\mathbb{R}^3} \frac{\partial}{\partial t} \sum w_i}_{\text{field energy}} \right) \stackrel{\text{local}}{=} \underbrace{- \int_{\mathbb{R}^2} \mathcal{S} \cdot \mathbf{n} \, df}_{\text{transported energy density} \quad \text{"skin"}}$$

Local Force Balance

Electro-Magnetic Force:

$$\dot{p} = F = q(\mathcal{E} + v \times \mathcal{B}) = \text{chargeforce} + \text{fieldforce} + \text{stresstension} \quad (40)$$

Charge Force:

$$\dots = \varrho \mathcal{E} + j \times \mathcal{B} \quad (41)$$

Field Force:

$$\dots = \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S} \quad (42)$$

Tension Force:

$$\nabla \cdot \mathcal{T} = \varepsilon_0 (\mathcal{E} \times \nabla \times \mathcal{E} - \mathcal{E} \nabla \cdot \mathcal{E}) + \frac{1}{\mu_0} (\mathcal{B} \nabla \times \mathcal{B} - \mathcal{B} \nabla \cdot \mathcal{B}) \quad (43)$$

Local Momentum Conservation:

$$\text{Force } F = q(\mathcal{E} + v \times \mathcal{B}) = \frac{d}{dt} p = 0$$

$$\text{Momentum } p = \int F = \frac{E}{c} = \text{const.}$$

$$\overbrace{\underbrace{\varrho \mathcal{E} + j \times \mathcal{B}}_{\text{charge force}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force}} + \underbrace{\nabla \cdot \mathcal{T}}_{\text{tension force}}}^{\dots} = 0 \quad (44)$$

Tension Force: Stress-Tensor

Stress Tensor quantifies tension $\hat{=}$ Forces = Pressure - Traction $\hat{=}$ “torsion tension” at angular momentum

$$\mathcal{T} \hat{=} \mathcal{T}_{ik} = \delta_{ik} \left(\frac{1}{2} \varepsilon_0 \mathcal{E}^2 + \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2 \right) - \varepsilon_0 \mathcal{E}_i \mathcal{E}_k - \frac{1}{\mu_0} \mathcal{B}_i \mathcal{B}_k \quad (45)$$

$\mathcal{E}$ and $\mathcal{B}$ fields of spacial confined charge and current density distribution, far field, static:

$$\mathcal{T} \sim \frac{1}{r^4} \Leftrightarrow \int_{\mathbb{R}^3} \mathcal{T} d^3 r = \int_{\infty} \mathcal{T} df = 0$$

Far Field: Force = 0 $\Leftrightarrow$ Momentum = const. $\Rightarrow$ Momentum Conservation

$$F = \dot{p} \hat{=} \frac{d}{dt} \underbrace{\left[\underbrace{\int_{\text{local}} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right]}_{=p=\text{const.}} = 0$$

Near Field (local): Force = Tension

$$F = \dot{p} \hat{=} \frac{d}{dt} \left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right] \stackrel{local}{=} - \underbrace{\int_{\mathbb{R}^2} \mathcal{T} \cdot \mathbf{n}}_{\text{stress tension appears}} \underbrace{df}_{\text{"skin"}}$$

$= p \neq const., \exists -\mathcal{T} \text{ local tension}$

Momentum Balance

Momentum Conservation and Local Tension: linear momentum conservation if $p = const.$, otherwise “tension” occurs:

$$p = \int F = \int \frac{d}{dt} p = \sqrt{\frac{1}{c^2} (E^2 - (mc)^2)} \stackrel{vacuum}{=} \frac{E}{c} = \frac{2\pi}{c^2} h\nu$$

$p = const.$, linear, vacuum, Far Field:

$$F = \dot{p} \hat{=} \int_{\mathbb{R}^3} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force density}} \right] = \frac{d}{dt} \left[\underbrace{\int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right] = 0$$

$= p \text{ momentum conserved if } p = const.$

(46)

$p \neq const.$, local $\in \mathbb{R}^2 \ll \mathbb{R}^3 = \infty$, linear, vacuum, Near Field $\Leftrightarrow \exists -\mathcal{T}$ as tension compensation flux:

$$F = \dot{p} \hat{=} \int_{local} \frac{d}{dt} \left[\underbrace{(\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge force density}} + \underbrace{\frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}}_{\text{field force density}} \right] = \frac{d}{dt} \left[\underbrace{\int_{local} (\varrho \mathcal{E} + j \times \mathcal{B})}_{\text{charge momentum}} + \underbrace{\frac{1}{c^2} \mathcal{S}}_{\text{field momentum}} \right] \stackrel{local}{=} - \int_{\mathbb{R}^2} \mathcal{T} \cdot \mathbf{n} \underbrace{df}_{\text{"skin"}}$$

$= p \neq const., \exists -\mathcal{T} \text{ local tension}$

(47)

Momentum $p = \int_{\mathbb{R}^3} (\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{1}{c^2} \mathcal{S}$

photon Spin $= \int_{\mathbb{R}^3} \frac{1}{c^2} \mathcal{S}$

angular momentum (photon Spin):

$$D = \dot{L} \hat{=}\dots \quad (48)$$

Radiation Pressure

average energy density $\hat{=}$ radiation pressure, if stationary (very slow flow motion $v \ll c$) :

$$\bar{w} \hat{=} -\nabla \mathcal{T} \cdot \mathbf{n} \hat{=} -\frac{1}{c} |\overline{\mathcal{E} \times \mathcal{H}}| = -\frac{1}{c} |\overline{\mathcal{S}}| = \overline{(\varrho \mathcal{E} + j \times \mathcal{B}) + \frac{\partial}{\partial t} \frac{1}{c^2} \mathcal{S}} \stackrel{\text{stationary}}{=} \overline{\varrho \mathcal{E} + j \times \mathcal{B}} \quad (49)$$

Abbreviations:

w_i = energy densities (electric w_e , magnetic w_m , gravitational, etc. ...), $i \in \mathbb{N}$

$\mathcal{S}_{w_i}$ = Poynting-Vectors $\hat{=}$ energy transport densities (transfers) by fields (electric, magnetic, gravitational, etc. ...) to corresponding energy densities w_i

$j \cdot \mathcal{E}$ = energy gain provided by fields $\mathcal{E}$ and $\mathcal{B}$ through current density j

$\mathcal{T}$ = Stress Tensor, (tension, momentum flow density)

$\mathbf{n}$ = surface normal vector (orthogonal on skin)

$\Delta \cdot := \text{div grad} = \text{Laplacian}$

J = “ignition of inhomogeneity, cause”

j = Current density

ϱ = Charge density

V = Potential difference

I = Current

R = Resistance

L = Inductance

C = Capacitance

Q = charge (ion)

$\Psi = LI = \text{magn. Flux}$

$\mathcal{S} = \mathcal{E} \times \mathcal{H} = \text{Poynting-Vector, radiation intensity}$

$D_S = \int (\cdot \mathcal{S}) dr = \int \text{div}(\mathcal{E} \times \mathcal{H}) dr = \text{dissipation energy lost from radiation} \approx 0$ for low frequencies ω

$d = \sqrt{\frac{1}{\mu\sigma} \frac{2}{\omega}}$ = skin depth (surface) of conductor

$r = \frac{d}{\sqrt{2}i} \rho = \frac{d}{1+i} \rho$ = radius (extension) of conductor

$\rho = kr$ = “radius of curvature k ”

μ = permeability of conductor

σ = conductivity of conductor (dielectricum)

Electromagnetic Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Vacuum

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Mechanical Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Vaccum

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Thermal Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Vacuum

. - Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast) - Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Gravitational Energy Exchange in Matter

- Near Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)
- Far Field, linear vs. angular, time independent (static/rest, stationary/slow) vs. time dependent (dynamic, fast)

...

Localization and Fields of Energy Transport

Localization of energy density w and of its energy transport $\mathcal{S}_w$ is where field force has strongest density. Fields for Forces, Energy, Momentum, Matter:

- Electromagnetic:

$$\left. \begin{array}{l} \text{Electric: } \mathcal{E}, \mathcal{D}, \mathcal{P}, \Phi_e \\ \text{Magnetic: } \mathcal{B}, \mathcal{H}, \mathcal{M}, \mathcal{A}, \Phi_m \end{array} \right\} \mathcal{S}, \mathcal{T}$$

- Thermal (skalar): T
- Gravitational: G

Frequency Cause-Effect

Frequency (commonness, probability, likelihood)

$$\omega = \frac{d}{dt}\varphi = \eta kv = kc = \frac{2\pi}{\mathcal{T}} = 2\pi \cdot \nu = \frac{a}{v} = \frac{v}{r} = \frac{E}{\hbar} = \sqrt{\frac{k}{m}} = \sqrt{\frac{r \times F}{m}} = \sqrt{\frac{g}{l}} \stackrel{\text{electromagnetic}}{=} \dots = \sqrt{\frac{1}{LC}} \stackrel{QM}{=} \frac{1}{2} \frac{p^2}{m \hbar} \quad (50)$$

$$\nu = \frac{E}{h} = \frac{v_{phs}}{\lambda} = \frac{c^2}{v} \frac{1}{\lambda} = \frac{\omega}{k} \frac{1}{\lambda} = \frac{\omega}{2\pi} \quad (51)$$

$$\nu_i = \frac{eV_{0i}}{h} + \nu_0 = \frac{E_{kin}}{\hbar} \omega \mathcal{T} + \nu_0 \quad (52)$$

$$\begin{array}{l} e = \text{electron charge} \quad V_0 = \text{Opposing Potential } (I = 0) \\ \text{Energy Level: } i = [1, n] \in \mathbb{N} \end{array} \quad (53)$$

$$E_{kin}^{max} = eV_0 = h(\nu - \nu_0) \quad (54)$$

Frequency contextual Causes

Frequency of rotational velocity as periodic occurrence of an event/state by space distance and time periodicity as well as movement speed and reach length, $v^2 = v_{ph} v_{gr}$, $\omega t = 2\pi \nu t = \frac{\lambda}{\lambda} \neq 0$:

$$\frac{v}{r} = \omega = \frac{\lambda}{t} \quad (55)$$

Rest state of source charge

Statics localization of charge. Only spacial demependency, no time dependency.

Energy transport ... Information transmission ...

$$v = 0 \quad \dot{v} = 0$$

Slow Movement of source charge

Stationarity localization of current flow by slow movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v = \text{const.} \ll c \quad \dot{v} = 0$$

Fast Movement of source charge

Fields forces by fast movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \approx c \quad \dot{v} = 0$$

Acceleration Movement of source charge

Radiation energy by accelerated movement of charge. Dynamics, time dependency.

Energy transport ... Information transmission ...

$$v \neq 0 \quad \dot{v} \neq 0$$

Movement in Vacuum

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{ph}} = 1 \quad 2\pi = k\lambda \quad v_{ph} = c \quad \omega = kc$$

$$\varepsilon_0 \mu_0 = \frac{1}{c^2}$$

$$\text{Continuity: } \nabla \cdot j = i\omega \varrho \quad i\omega Q = 0$$

Movement in Matter

$$\eta = \frac{\tan \alpha}{\tan \beta} = \frac{c}{v_{Ph}} = \sqrt{\frac{\varepsilon \mu}{\varepsilon_0 \mu_0}} \neq 1 \quad v_{ph} = \pm \sqrt{\frac{1}{\varepsilon \mu} - \frac{\sigma^2}{4\varepsilon^2 k^2}} \quad \omega = kv_{ph} = kc \sqrt{\frac{\varepsilon_0 \mu_0}{\varepsilon \mu}} =$$
$$kc \frac{\tan \beta}{\tan \alpha} = k \frac{c}{\eta}$$

$$\varepsilon \mu = \left(\frac{\eta}{c}\right)^2$$

$$\text{Dispersion: } k^2 = \varepsilon \mu \omega^2 + i \mu \sigma \omega$$

Near Field

$$r_0 \ll r \ll \lambda \quad kr \ll 2\pi \quad \frac{w_m}{w_e} \ll 1$$

r is small, fast movement

Far Field

$$r \gg \lambda \quad kr \gg 1 \quad r \gg kr_0^2 \ll 1 \quad \frac{w_m}{w_e} = 1$$

r is big, slow movement

High Frequency

$$\omega \gg \frac{2}{\mu \sigma r_0^2}$$

λ is big, ν frequency is small, short time period

Low Frequency

$$\omega \ll \frac{2}{\mu \sigma r_0^2}$$

λ is small, ν frequency is big, long time period

Conditions

Border (skin surface): ...

Interior (medium body): ...

Global:

Local:

Consequences

dielectricity constant (matter medium): $\varepsilon = \varepsilon(\omega) = \mathcal{E}^{-1} \mathcal{D} = \mathcal{E}^{-1} (\mathcal{P} + \varepsilon_0 \mathcal{B})$

permeability constant (matter medium): $\mu = \mu(\omega) = \mathcal{B}^{-1} \mathcal{H} = \mathcal{B}^{-1} \left(\frac{1}{\mu_0} \mathcal{B} - \mathcal{M} \right)$

conductivity (matter medium): $\sigma = \sigma(\omega) = \frac{\sigma_0}{\sqrt{1+\omega^2 t^2}}$

magn. energy density (vacuum): $w_m = \frac{1}{2} \frac{1}{\mu_0} \mathcal{B}^2$

elec. energy density (vacuum): $w_e = \frac{1}{2} \varepsilon_0 \mathcal{E}^2$

wave vector: $k = \frac{2\pi}{\lambda} = \eta \frac{\omega}{c}$

refraction index (matter medium): $\eta = \frac{\tan \alpha}{\tan \beta} = \frac{\eta_{medium1}}{\eta_{medium2}}$

Energy: $E = h\nu = \hbar\omega = E_{kin} + E_{pot} \stackrel{!}{=} \sqrt{(pc)^2 + (mc^2)^2}$

Conditions (skin, interface, border, surface)

Boundary Conditions

$$\mathbf{n} \cdot [\mathcal{D}] = \sigma$$

$$\mathbf{n} \cdot [\mathcal{B}] = 0$$

$$\mathbf{n} \times [\mathcal{E}] = 0$$

$$\mathbf{n} \times [\mathcal{H}] = J$$

Interior Conditions

e.g. for Superconductivity

$$\mathbf{n} \cdot \mathcal{H} = 0$$

$$\mathbf{n} \times \mathcal{D} = 0$$

Abbreviations

$\mathbf{n}$ = normal vector (orthogonality orientation)

$\mathcal{E}$ = elec. Field (“traction force”?)

$\mathcal{B}$ = magn. Field (“compressive force”?)

$\mathcal{D}$ = elec. Displacement Field (“exitioan”)

$\mathcal{H}$ = magn. Exitation Field (“incentivation”)

σ = electric charge at surface

J = current density at surface

Distinguishing Situations Conditions and Dependencies

effects of charge (separaton) and differentiating situations (conditioning frameworks) for fields and potentials:

Spacetime Embedding

- Spacial dependence (mechanics)
- Time dependence (dynamics)

Movement Framework

- Static: rest system, equilibrium
- Dynamic:
 - Stationary: slow motion
 - Relativistic: fast motion
 - Radiation; accelerated motion

Medium Property

- Vacuum: spacial mean = time mean
- Matter:
 - Mass (inertia)
 - Charge (conductor, ion)
 - Dielectricum (isolator)
 - Dispersion (refraction)

Field Distance

- Near Field (granular spacetime, mean average values)
- Far Field (flat spacetime)

Localisation Identity

- Global (regular)
- Local (singular)

-> Tool concepts: Momentum, Force, Energy

-> Conditions framework: - continuity condition (charge, current) - equilibrium conditions (action vs. reaction, homogeneity vs. inhomogeneity) - initial conditions (starting point) - border (skin) conditions (differentiation) - interior conditions (integration) - conservations of local densities (charge, energy, momentum)

-> Properties: - characteristics from charge: dielectricity ε , permeability μ , conductivity σ , frequency ω - characteristics from heat: ... - Properties from gravitation: ...

Operators

$\nabla := \text{grad} = \text{Gradient (Gefälle, Neigung)}$

$\nabla \cdot := \text{div} = \text{Divergence (Quelle)}$

$\nabla \times := \text{rot} = \text{Rotation (Wirbelung, Curl)}$

$\nabla^2 = \text{Lagrange Operator (Krümmung, Beschleunigung)}$

$\Delta \cdot := \text{div(grad)} = \text{Laplace Operator}$

$\Delta := \text{Difference, shift}$